

Hands-On Learning Moment for Layout Creation and Verification Using NGSPICE

Task: Create and verify a layout for
a basic logic gate

Scenario

You are a chip design enthusiast learning to use open-source EDA tools for creating digital layouts. Your task is to design a layout for a basic logic gate (e.g., a NAND gate) using Magic and verify its functionality using NGSPICE.

Instructions

Step 1: Create the Logic Gate Layout in Magic

1. Launch Magic:

Open the Magic tool on your system by typing magic in the terminal.

2. Set Up Your Workspace:

→ Open the workspace folder where you want to save your layout files.

→ Load the technology file for your process using the command:

```
tech load <technology_file>
```

3. Design the Layout:

→ Create a simple NAND gate layout:

Use **nMOS** and **pMOS** transistors.

Include necessary layers for diffusion, polysilicon, and metal connections.

→ Ensure the input and output pads are clearly labeled (e.g., A, B, and OUT).

4. Verify the Layout:

→ Run design rule checks (DRC) using:

drc check

→ Save the layout:

save nand_gate.mag

Step 2: Extract Netlist from Magic

1. Generate SPICE Netlist:

→ Extract the SPICE netlist using the command:

extract all

ext2spice

2. Edit the SPICE Netlist:

→ Open the .sp file generated and add SPICE simulation parameters, such as voltage sources and simulation controls.

Step 3: Simulate in NGSPICE

1. Open NGSPICE:

→ Launch NGSPICE by typing ngspice in the terminal.

2. Load the Netlist:

→ Load your SPICE netlist into NGSPICE:

```
source nand_gate.sp
```

3. Run the Stimulation:

→ Add input stimulus and run the simulation:

```
tran 1n 1n
```

→ View output waveforms to verify the correct behavior of the NAND gate.

Step 4: Analyze and Validate Results

1. Check Simulation Outputs:

→ Use the plot command to view waveforms:

```
plot V(out)
```

→ Verify that the output corresponds to the expected NAND gate logic.

2. Troubleshoot if Necessary:

- Modify the layout or netlist based on simulation results.
- Re-run simulations until the output meets specifications.

This activity introduces key concepts in chip layout design and verification. It leverages Magic and NGSPICE for practical learning. Complete the process to solidify your understanding of these powerful EDA tools.